

BROADVIEW, SK, Canada

WMO: 718610

Lat: **50.368N**Lon: **102.571W**Elev: **1968**StdP: **13.68**Time Zone: **-6.00 (W06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.9	-21.9	-34.7	0.8	-26.7	-29.5	1.1	-21.6	29.4	18.2	26.4	16.8	7.6	300	0.705

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	84.6	66.6	81.3	65.2	78.4	63.6	70.1	79.9	67.6	77.6	65.6	75.5	11.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	105.6	75.9	64.0	96.1	73.1	61.8	88.7	70.5	35.4	80.1	33.2	77.4	31.5	75.7	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	26.4	22.9	20.3	DB	-34.3	91.5	5.7	3.4	-38.4	94.0	-41.8	96.0	-44.9	97.9	-49.1	100.4
(5)				WB	-34.6	74.3	5.6	2.7	-38.6	76.2	-41.9	77.7	-45.0	79.2	-49.0	81.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	5.4	8.2	20.4	37.7	50.3	59.8	64.9	62.5	53.8	39.5	23.3	10.0
	DBStd	23.97	15.69	14.56	14.30	10.93	8.42	6.37	5.49	6.17	8.11	9.23	12.41	14.64
	HDD50	6391	1382	1169	917	385	100	5	0	1	50	342	801	1239
	HDD65	10554	1847	1589	1382	819	459	177	71	122	343	790	1251	1704
	CDD50	1460	0	0	0	16	110	300	461	389	166	18	0	0
	CDD65	146	0	0	0	0	4	22	67	45	8	0	0	0
	CDH74	1850	0	0	0	5	96	260	698	587	198	6	0	0
	CDH80	422	0	0	0	0	15	48	158	161	39	1	0	0
Wind	WSAvg	9.5	9.8	9.7	10.2	10.3	10.4	9.0	8.1	7.9	9.0	9.7	9.8	9.6
Precipitation	PrecAvg	16.4	0.7	0.5	0.9	1.1	1.9	2.4	2.2	2.2	2.2	1.1	0.6	0.7
	PrecMax	21.7	1.5	1.3	2.3	4.3	4.8	5.2	4.5	6.1	5.7	2.6	1.4	1.3
	PrecMin	11.6	0.1	0.2	0.2	0.0	0.3	0.3	0.5	0.3	0.1	0.0	0.1	0.3
	PrecStd	2.4	0.4	0.3	0.5	0.9	1.2	1.3	1.1	1.5	1.5	0.9	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	39.8	56.4	73.2	82.1	85.0	88.2	89.4	84.8	72.0	58.3	42.3
		MCWB	34.7	35.6	45.8	53.4	60.4	67.9	70.3	66.9	63.1	54.6	46.4	36.0
	2%	DB	35.8	35.8	48.4	65.6	76.3	80.4	84.4	84.4	79.8	64.9	48.9	37.9
		MCWB	32.6	32.5	40.4	49.7	56.6	63.6	68.6	65.9	61.2	51.4	40.7	33.6
	5%	DB	32.0	32.5	42.3	60.8	71.3	76.6	80.9	80.4	74.9	58.9	43.5	34.1
		MCWB	29.6	30.3	37.4	46.6	54.1	61.9	67.0	65.1	59.2	48.2	37.5	31.3
	10%	DB	27.8	29.1	37.8	55.5	66.8	73.2	78.0	76.5	69.7	53.9	39.0	29.8
		MCWB	26.1	27.2	34.4	44.3	51.9	60.1	65.4	62.9	56.6	46.0	35.1	27.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.2	36.0	46.1	54.5	62.2	72.2	74.0	71.7	66.8	57.0	46.8	36.7
		MCDB	38.7	39.2	54.6	69.6	73.3	80.3	83.1	82.5	78.9	68.8	57.4	41.6
	2%	WB	32.7	32.9	41.2	51.1	58.4	66.6	71.1	68.7	63.3	52.6	41.3	33.9
		MCDB	35.7	35.2	47.7	63.9	71.2	76.1	80.4	79.8	75.2	62.1	47.7	37.4
	5%	WB	29.8	30.3	37.5	47.7	55.9	64.0	68.7	66.4	60.4	49.4	38.0	31.1
		MCDB	32.0	32.5	41.7	58.5	68.2	73.2	78.1	77.4	72.2	57.1	42.6	34.0
	10%	WB	26.2	27.2	34.5	45.0	53.6	61.6	66.5	64.3	57.6	46.6	35.2	27.9
		MCDB	27.8	29.1	37.6	54.6	64.1	70.5	76.1	74.1	67.7	53.5	38.9	29.9
Mean Daily Temperature Range	5% DB	MDBR	17.3	17.6	17.7	20.5	22.5	19.7	21.2	23.1	23.0	18.9	16.1	16.2
		MCDBR	18.6	16.4	20.9	29.5	30.3	25.6	25.9	29.3	31.5	28.5	22.7	18.9
		MCWBR	16.7	14.0	14.8	16.1	14.4	12.4	13.5	14.4	16.0	16.8	15.5	15.8
	5% WB	MCDBR	18.3	16.0	20.3	27.5	25.7	21.7	23.3	26.0	28.1	25.8	21.1	18.6
		MCWBR	16.6	13.9	14.8	15.8	13.3	12.0	13.5	14.1	15.3	15.6	14.9	15.9
Clear-Sky Solar Irradiance	taub	0.261	0.249	0.274	0.332	0.340	0.358	0.367	0.382	0.341	0.282	0.247	0.225	
	taud	2.245	2.326	2.432	2.379	2.412	2.393	2.389	2.330	2.424	2.563	2.502	2.388	
	Ebn at Noon	246	288	299	289	291	285	280	269	271	274	260	253	
	Edh at Noon	22	26	29	35	35	36	36	36	30	21	17	16	
All-Sky Solar Radiation	RadAvg	412	744	1168	1553	1846	1900	2060	1636	1235	723	399	294	
	RadStd	34	70	88	100	154	141	115	128	105	89	59	24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A